

COMICCRAFT

AI Comic Story Creator using Gemini Models

PHASE 2: Requirement Analysis Phase

Functional Requirements

The system must collect the information needed to personalize comic generation and process it through the AI workflow.

User Input Requirements

- Story Prompt – the main idea or storyline provided by the user.
- Character Name – the main character of the comic.
- Setting – the location or environment such as forest, school, city, or space.
- Story Tone – the desired mood such as dramatic, funny, poetic, or light-hearted.
- Art Style – the preferred visual style such as anime, pixel art, comic book, or realistic.

Generation Requirements

- Generate a structured 5-panel comic outline.
- Generate detailed narration and character dialogues for the panels.
- Generate comic-style illustrations using image prompts.
- Organize the images and story text into a complete comic layout.

System / Technical Requirements

- FastAPI for backend routing and server-side processing.
- HTML, CSS, and Jinja2 for the web interface and dynamic templates.
- Google Gemini models for outline, narration, and dialogue generation.
- Hugging Face Diffusers with Stable Diffusion for image generation.
- FPDF for generating the downloadable comic PDF.
- Python environment with required packages and version-control support.

Expected Output

The final system should produce a personalized, visually engaging comic that can be previewed in the browser and exported as a PDF containing the generated images and narrative content.

Project documentation prepared for phase-wise submission.